

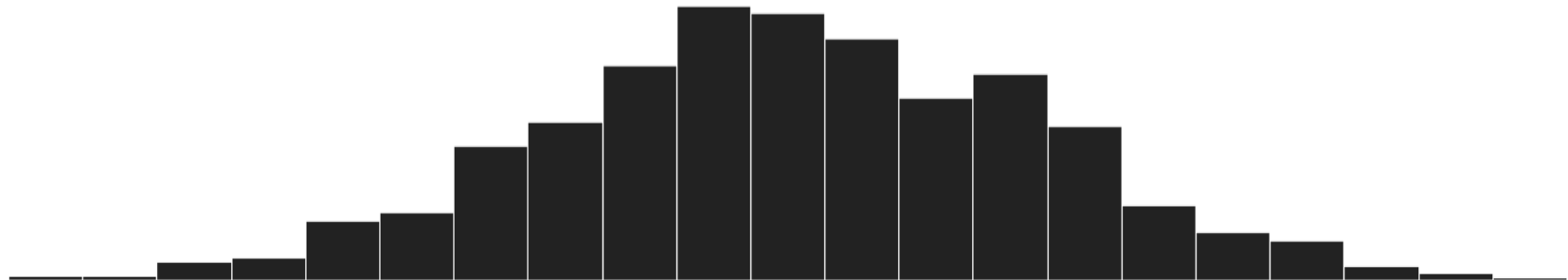
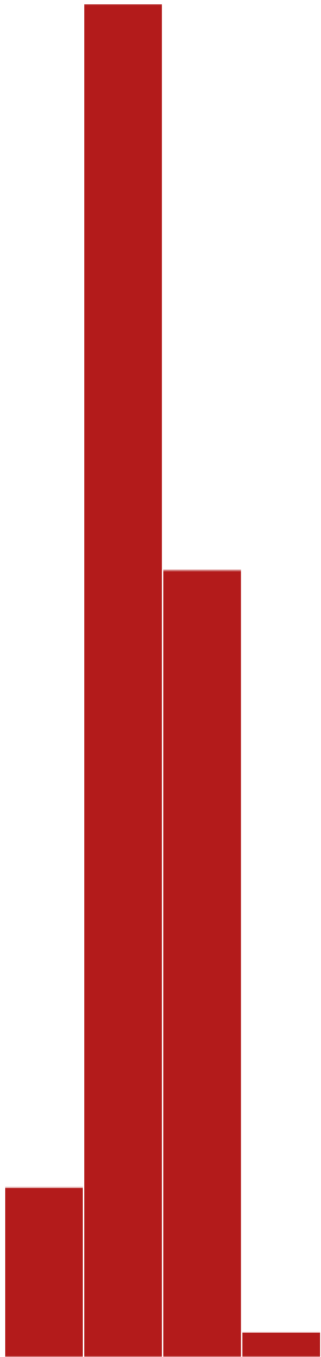
VTPEH 6105

Lecture 07

HYPOTHESIS TESTING: LINEAR REGRESSIONS

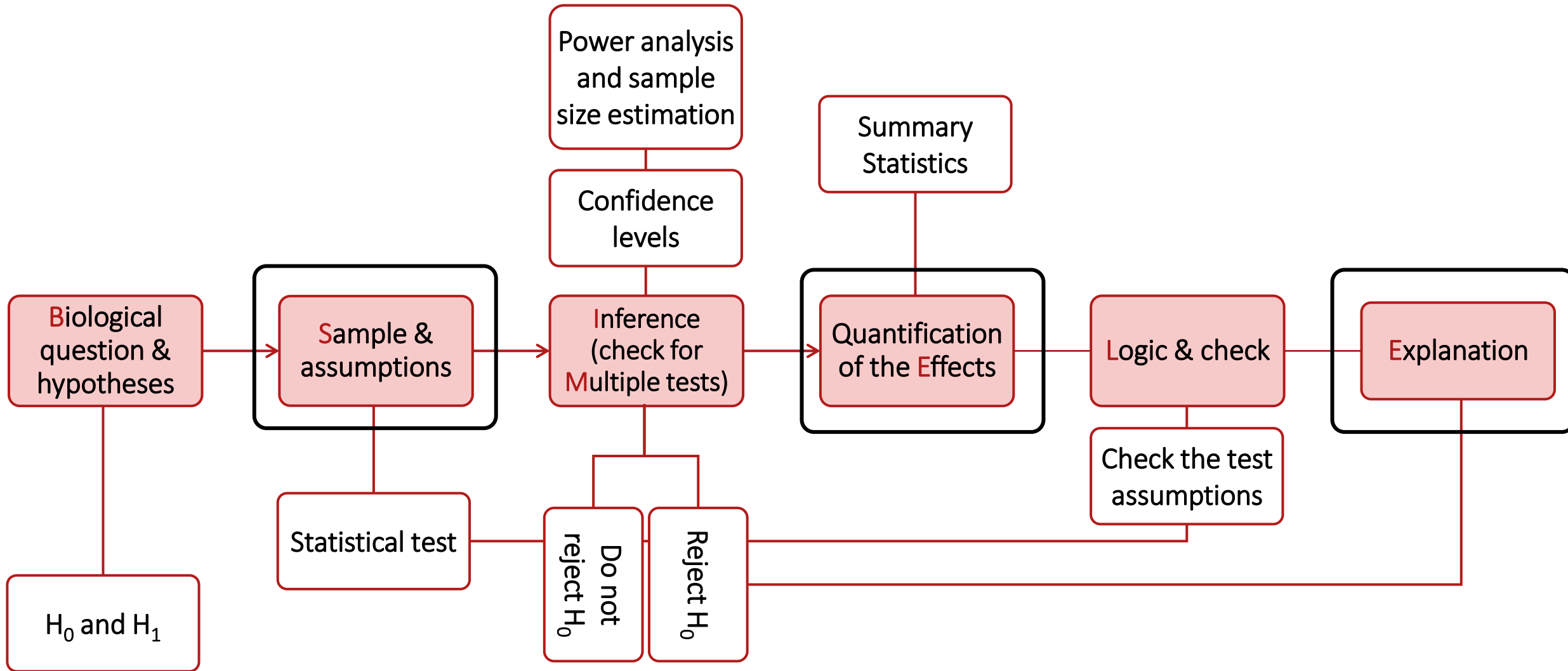
Amandine Gamble

amandine.gamble@cornell.edu



House Keeping

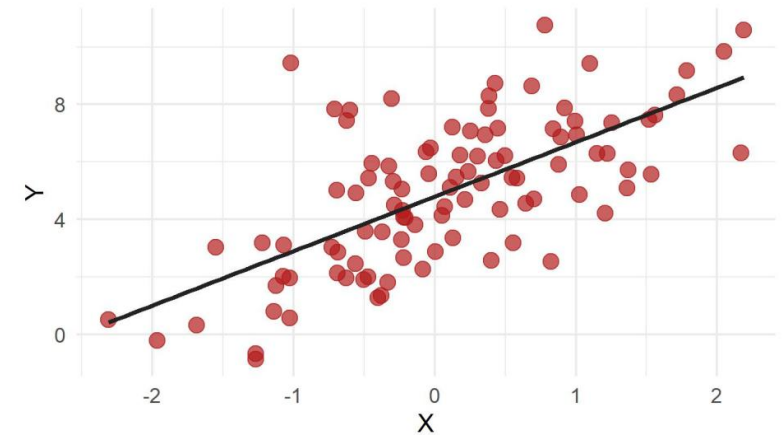
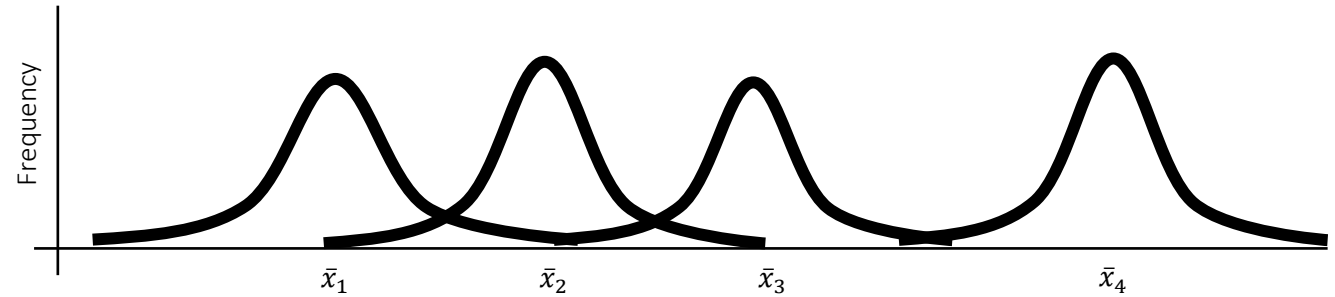
Feedback on Previous Homework: Key Missing Parts



Hypothesis Testing Refresher

What We've Seen so Far...

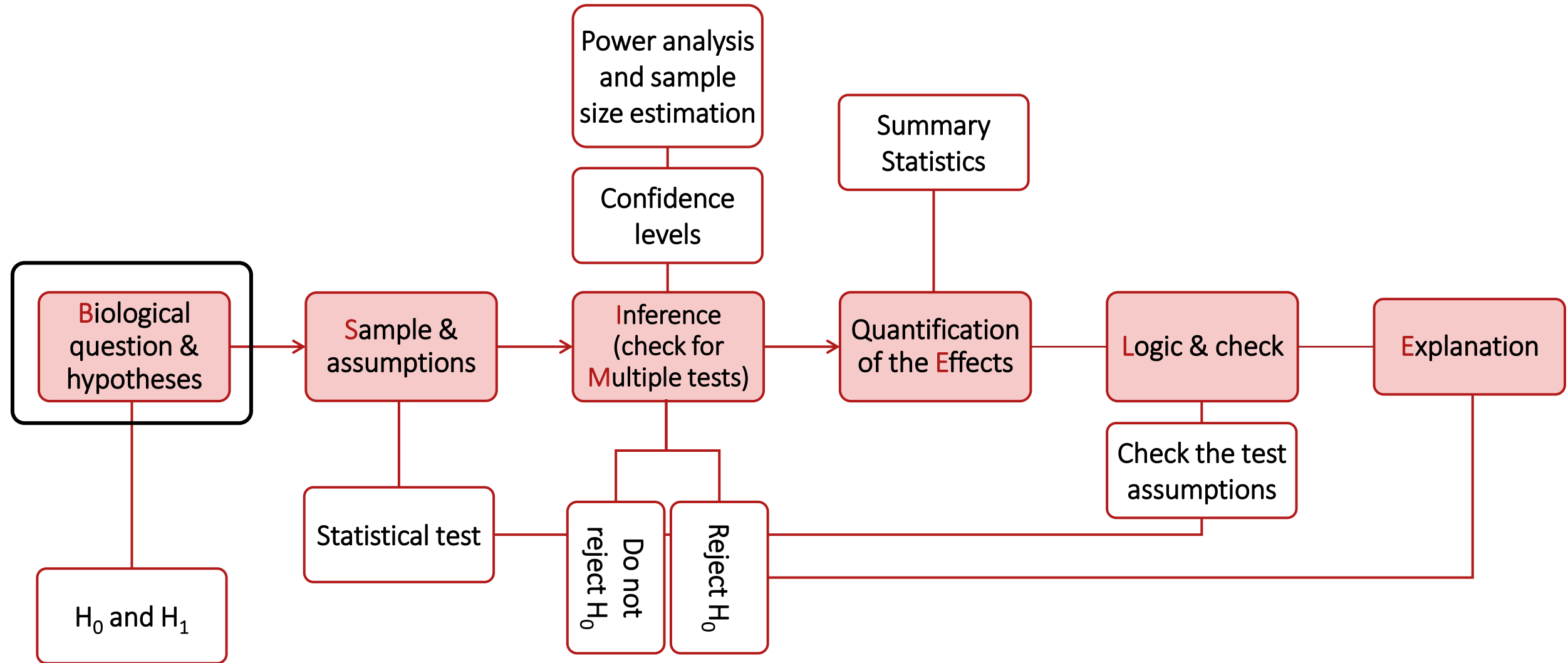
- Continuous variable ~ categorical variable
 - T-test
 - Wilcoxon
 - ANOVA
 - Kruskal-Wallis
- Categorical variable ~ categorical variable
 - Chi2
- Continuous variable ~ continuous variable
 - Pearson
 - Spearman
 - Linear regression model



Testing & Quantifying Associations between Continuous Variables

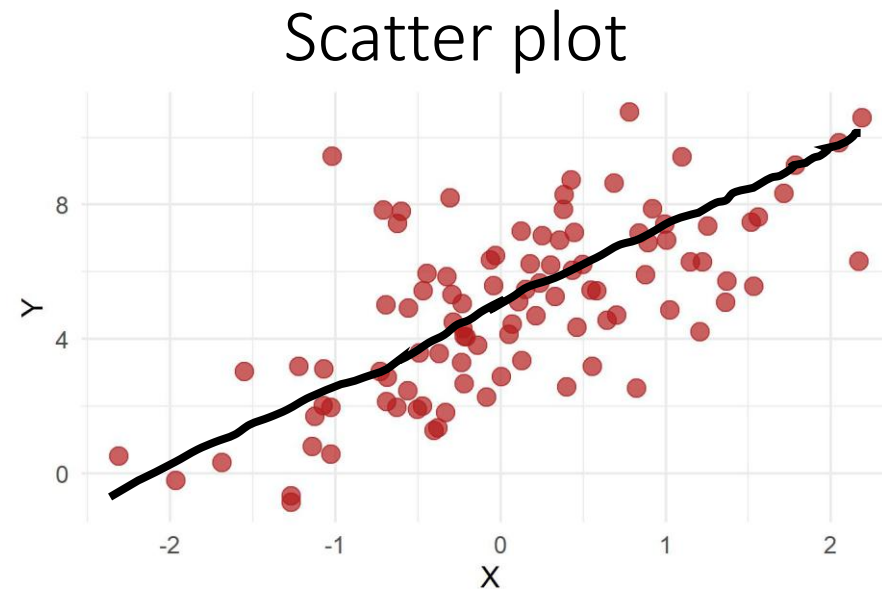
Hypothesis testing with correlation coefficients

Hypothesis Testing Workflow



Relationships Between Two Continuous Variables

- Visualization
- First assessment of linearity...

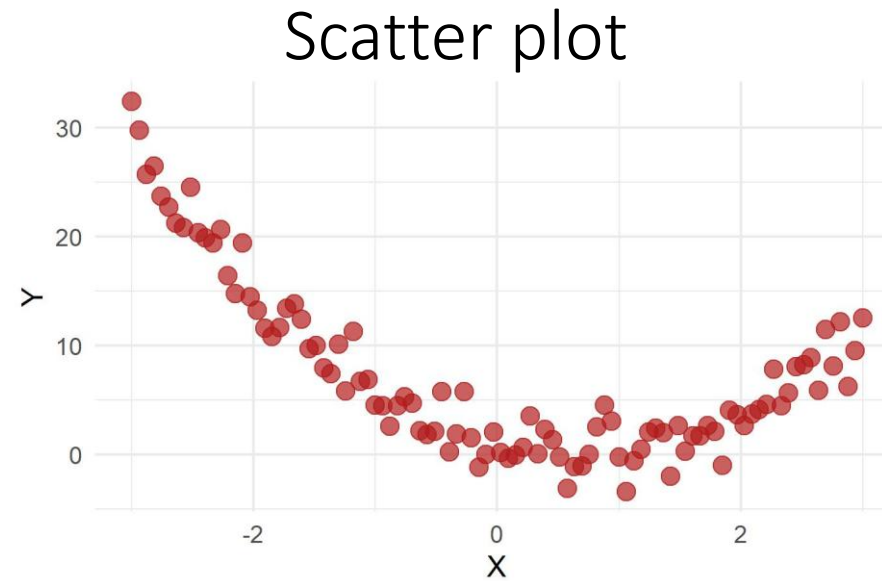


Looks kind of linear

I.e., drawing a line seem to capture the relationship

Relationships Between Two Continuous Variables

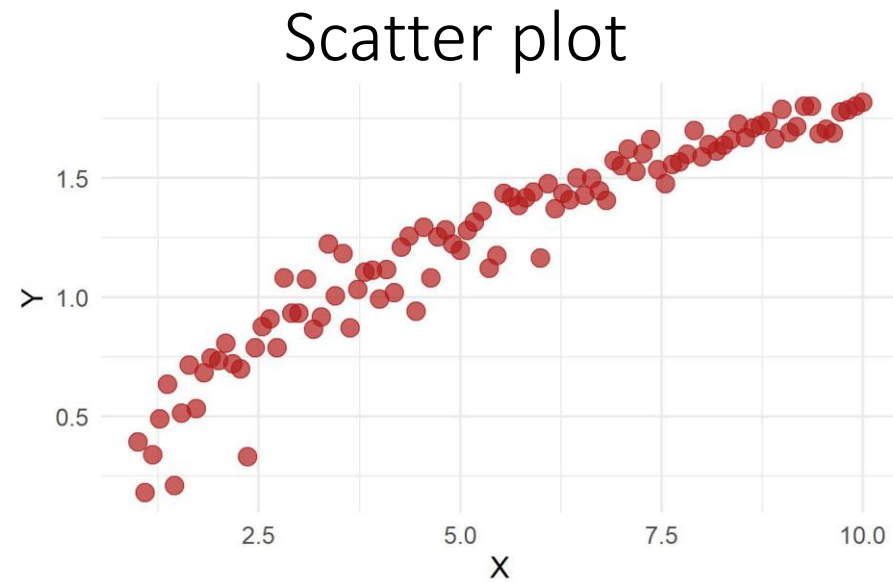
- Visualization
- First assessment of linearity...



Does not look linear
Different shape

Relationships Between Two Continuous Variables

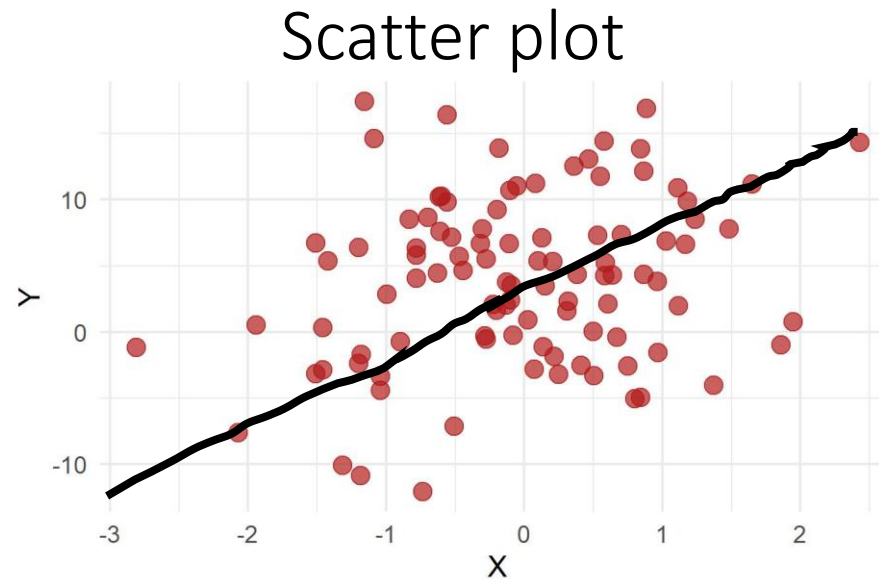
- Visualization
- First assessment of linearity...



Does not look linear
Different shape

Relationships Between Two Continuous Variables

- Visualization
- First assessment of linearity...

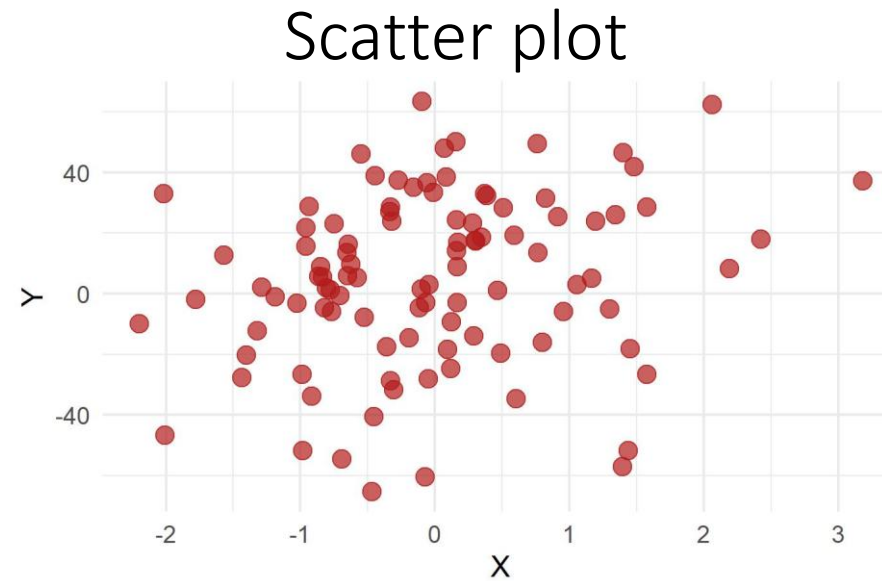


Maybe it could be linear?

I can roughly imagine a line, but the points are widely spread around it, it is noisy

Relationships Between Two Continuous Variables

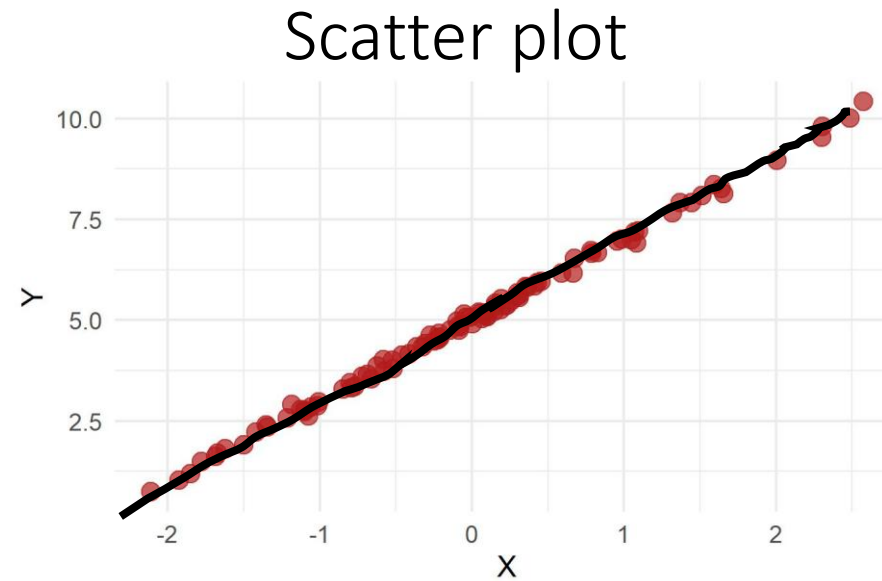
- Visualization
- First assessment of linearity...



Does not look linear
Too much noise

Relationships Between Two Continuous Variables

- Visualization
- First assessment of linearity...
- Look at
 - Shape
 - Noise (dispersion around the shape)
 - And it has to make sense!



This is clearly linear

Regressions: Formalization

- Regression: **estimation the relationships** between variables

We often talk about dependent variable Y (often called the 'outcome' or 'response' variable) and one or more independent variables X (often called 'predictors', 'covariates', 'explanatory variables')

$$Y \sim X \text{ or } Y \sim X_1, X_2, \dots$$

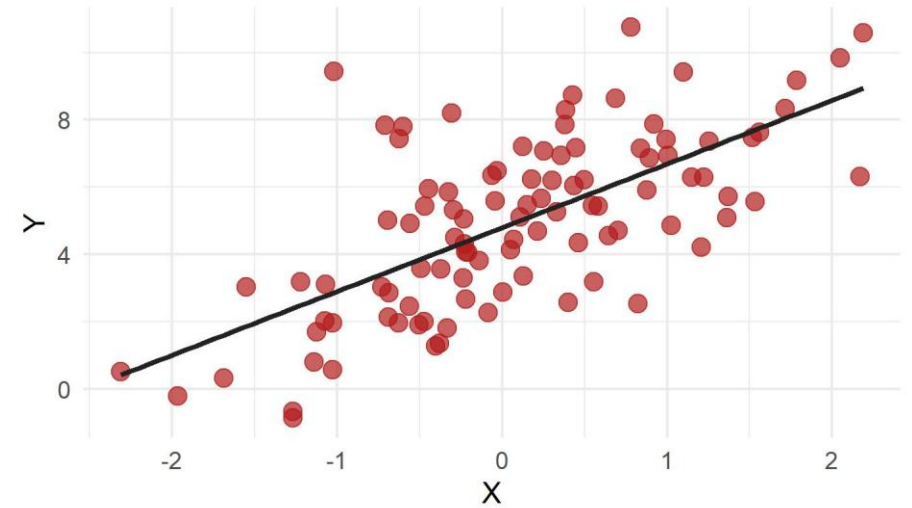
→ An objective way to **predict** or estimate the value of one variable, given a value of another variable

- **Linear** regression:

$$Y = aX + b \text{ and } y_i = \underbrace{ax_i + b}_{\text{Predicted value}} + \varepsilon_i$$

Individual variation, or noise: deviance from the predicted value

Regressions: Formalization



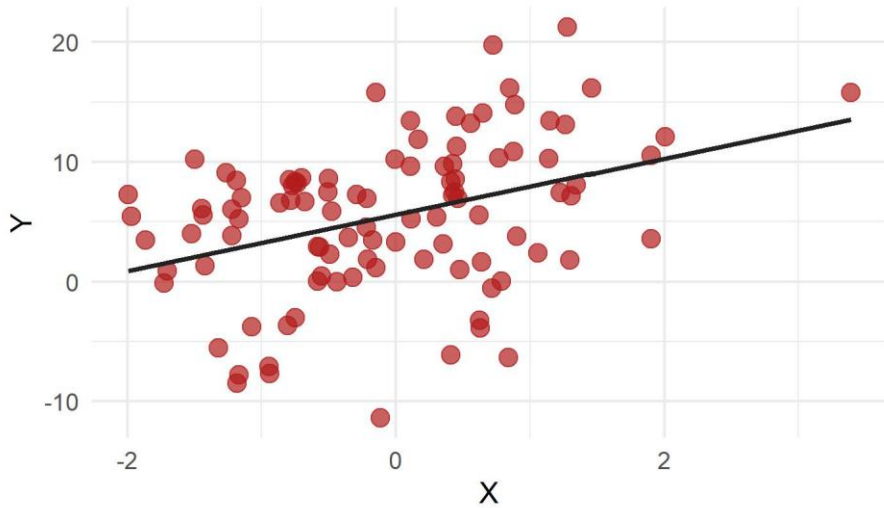
→ An objective way to **predict** or estimate the value of one variable, given a value of another variable

- **Linear** regression:

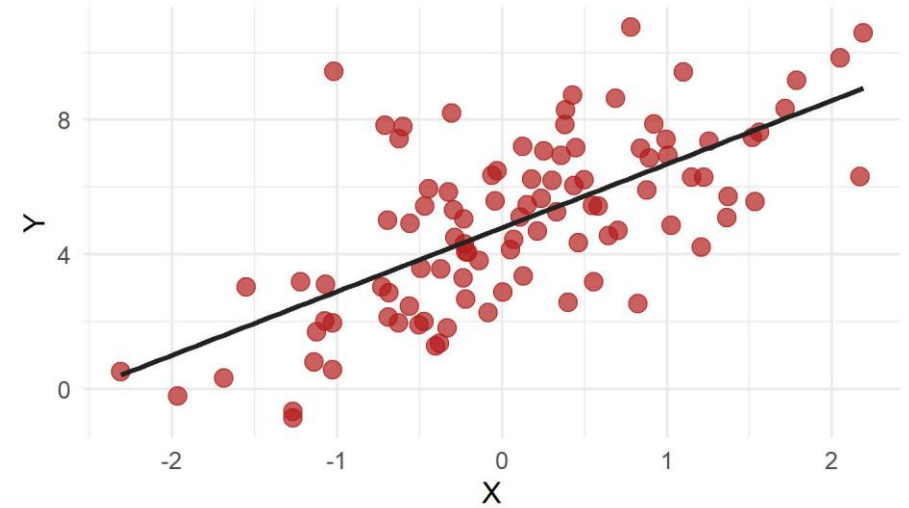
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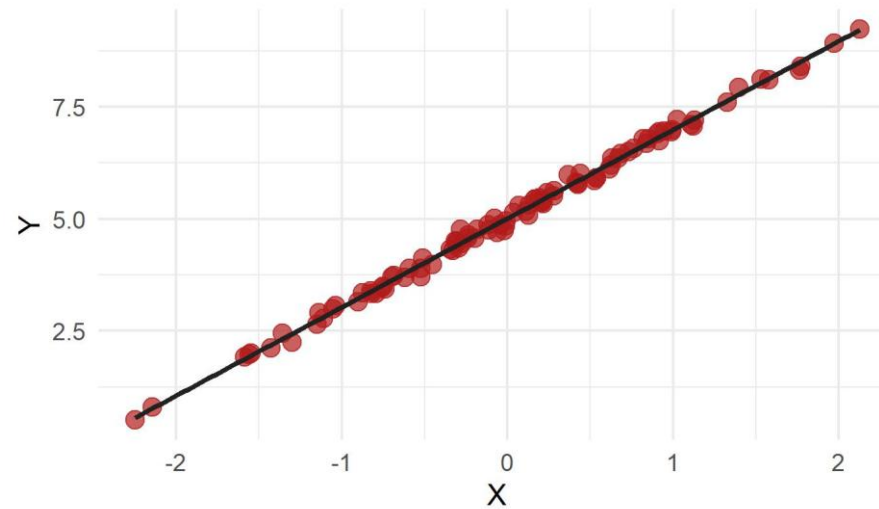
Regressions: Quantification with Correlation Coefficients



Weak positive
correlation



So so positive
correlation



Strong positive
correlation

Regressions: Quantification with Correlation Coefficients

- Correlation coefficients measure the **strength** and **direction** of association between continuous variables

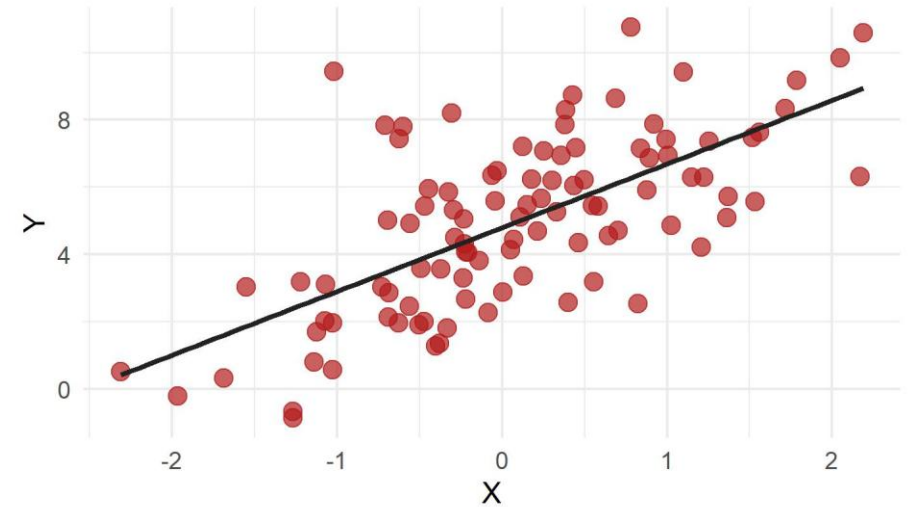
$$Coef \in [-1; 1]$$

$Coef \rightarrow 1$ for strong positive associations

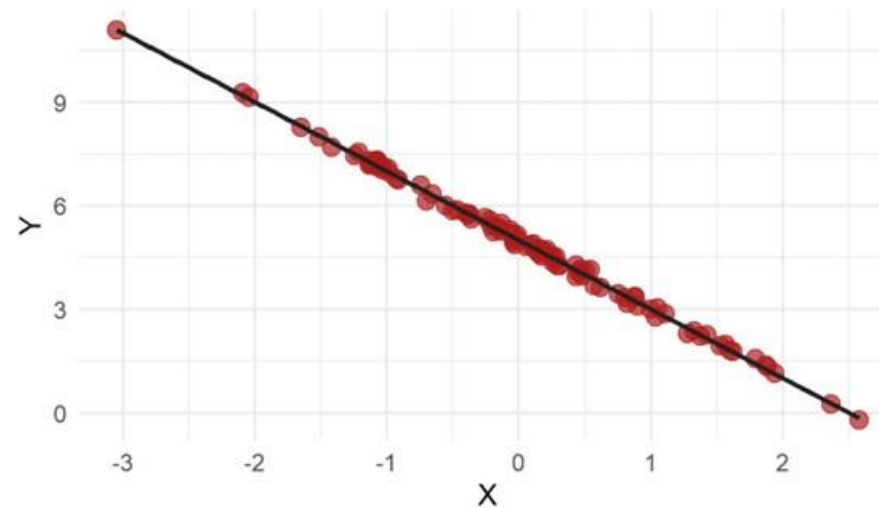
$Coef \rightarrow -1$ for strong negative associations

$Coef = 0$ in the absence of association

$Coef = 1$ for perfect associations



So so positive correlation



Strong negative correlation

Regressions: Quantification with Correlation Coefficients

- Correlation coefficients measure the **strength** and **direction** of association between continuous variables

$$Coef \in [-1; 1]$$

$Coef \rightarrow 1$ for strong positive associations

$Coef \rightarrow -1$ for strong negative associations

$Coef = 0$ in the absence of association

- Correlation coefficient are parameters

Like means and medians are central tendency parameters, and standard deviations and interquartile ranges are dispersion parameters... Except that correlation coefficients are measured over 2 variables

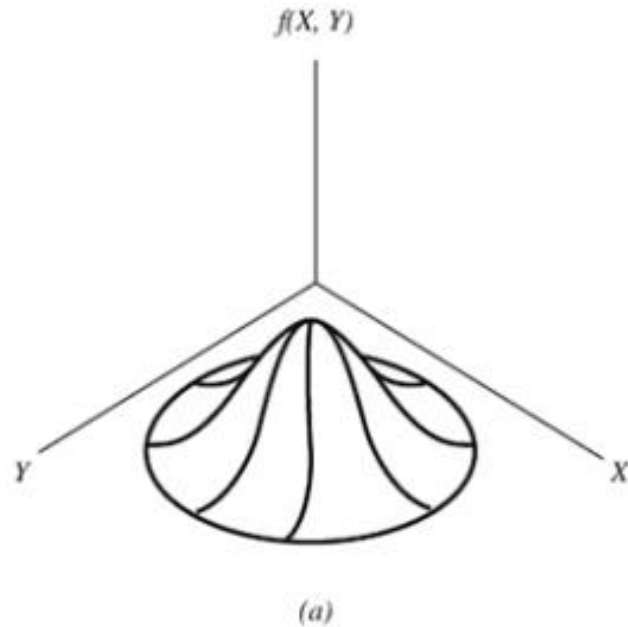
Like other parameters, there is a **true population correlation coefficient ρ** (rho) and an **observed sample correlation coefficient r**

How do we calculate r ?

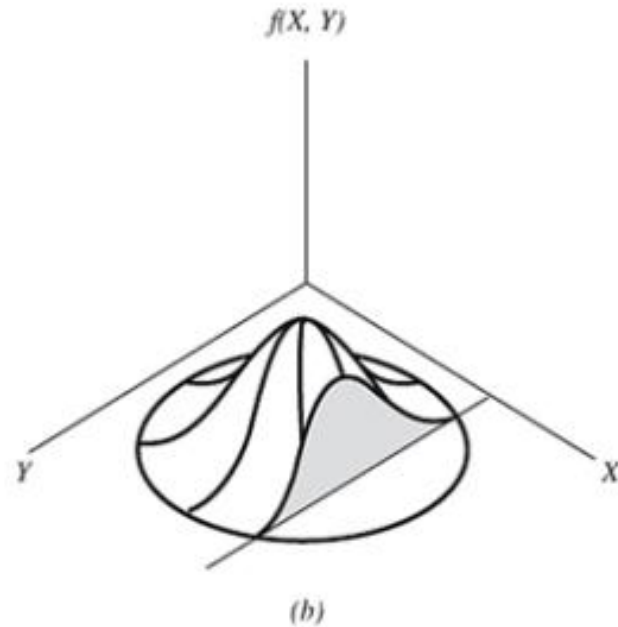
Depends on the assumptions on the data...

Pearson's Correlation Coefficient

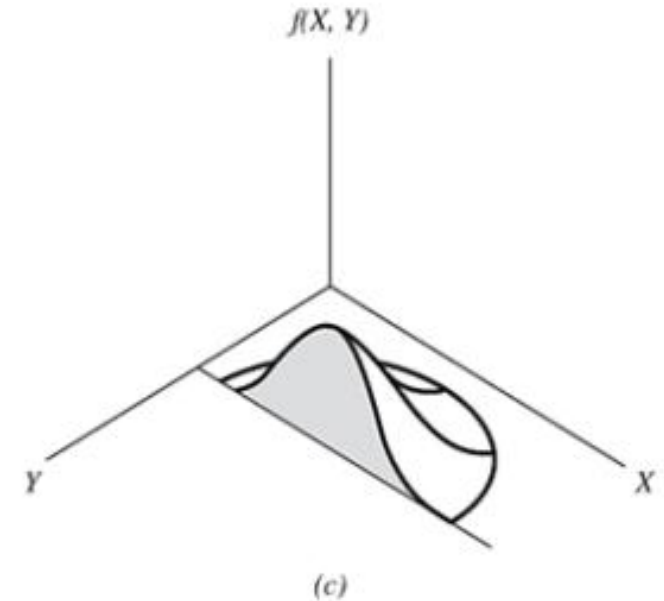
- The bivariate normal distribution of X and Y has five parameters: μ_x , μ_y , σ_x , σ_y and ρ with **ρ the population correlation coefficient**



A bivariate normal distribution.
Joint distribution of X and Y .

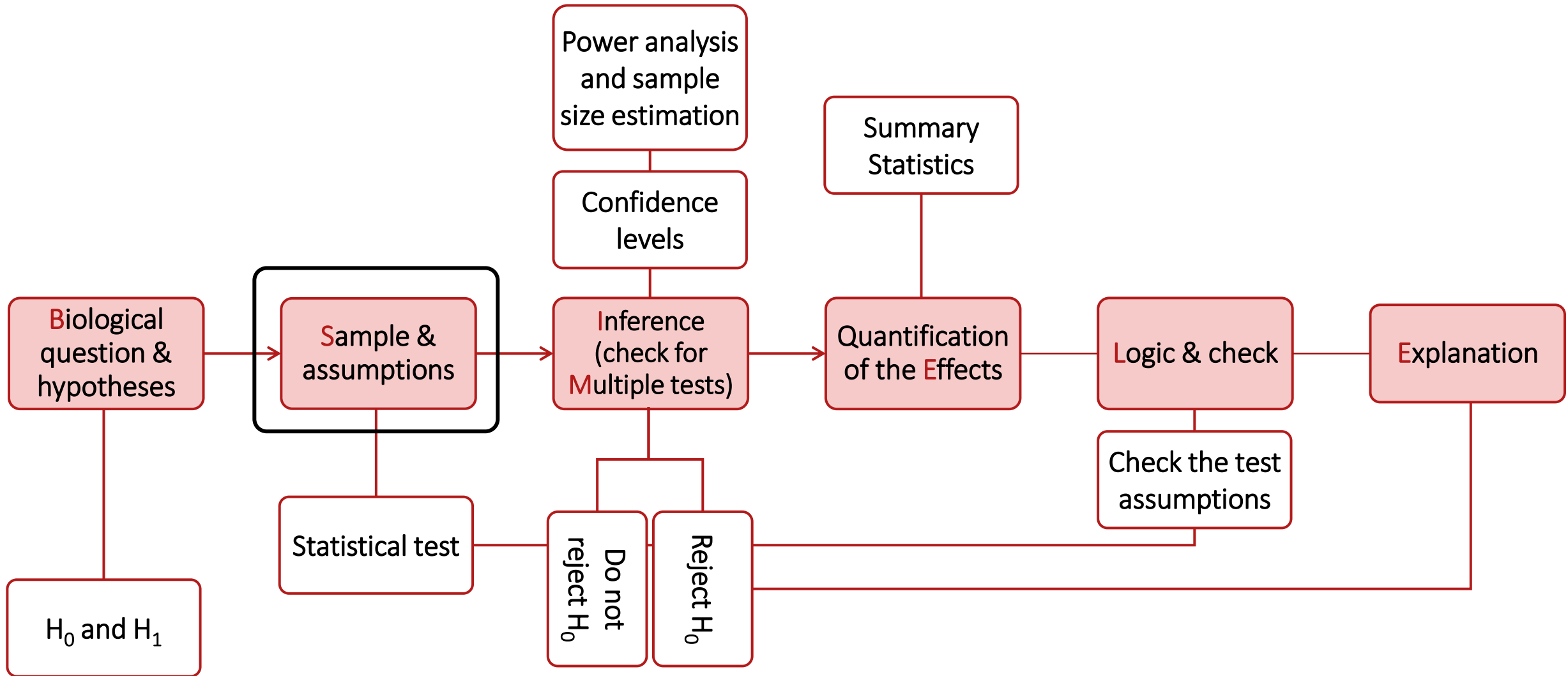


For each value of $Y|X$



For each value of $X|Y$

Hypothesis Testing Workflow



Pearson's Correlation Coefficient: Assumptions

- **Linearity between X and Y** : the means of the subpopulations of Y all lie on the same straight line defined by

$$\mu_{y|x} = a x + b$$

- **Independence**: Y values are statistically independent (via ε_i)

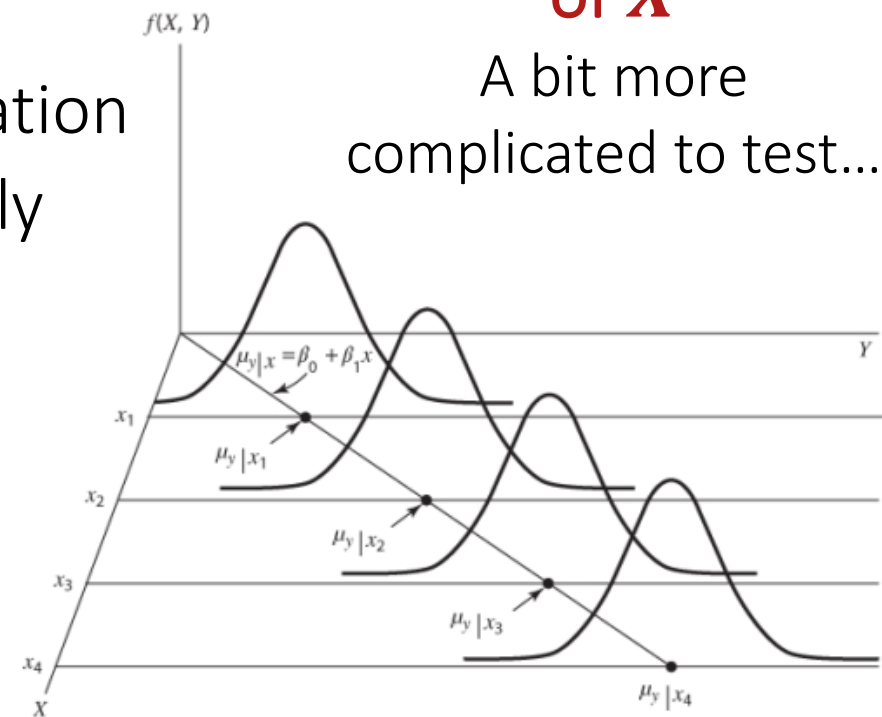
$$y_i = a x_i + b + \varepsilon_i$$

- **Normality**: for each value of X there is a subpopulation of Y values; these subpopulations must be normally distributed

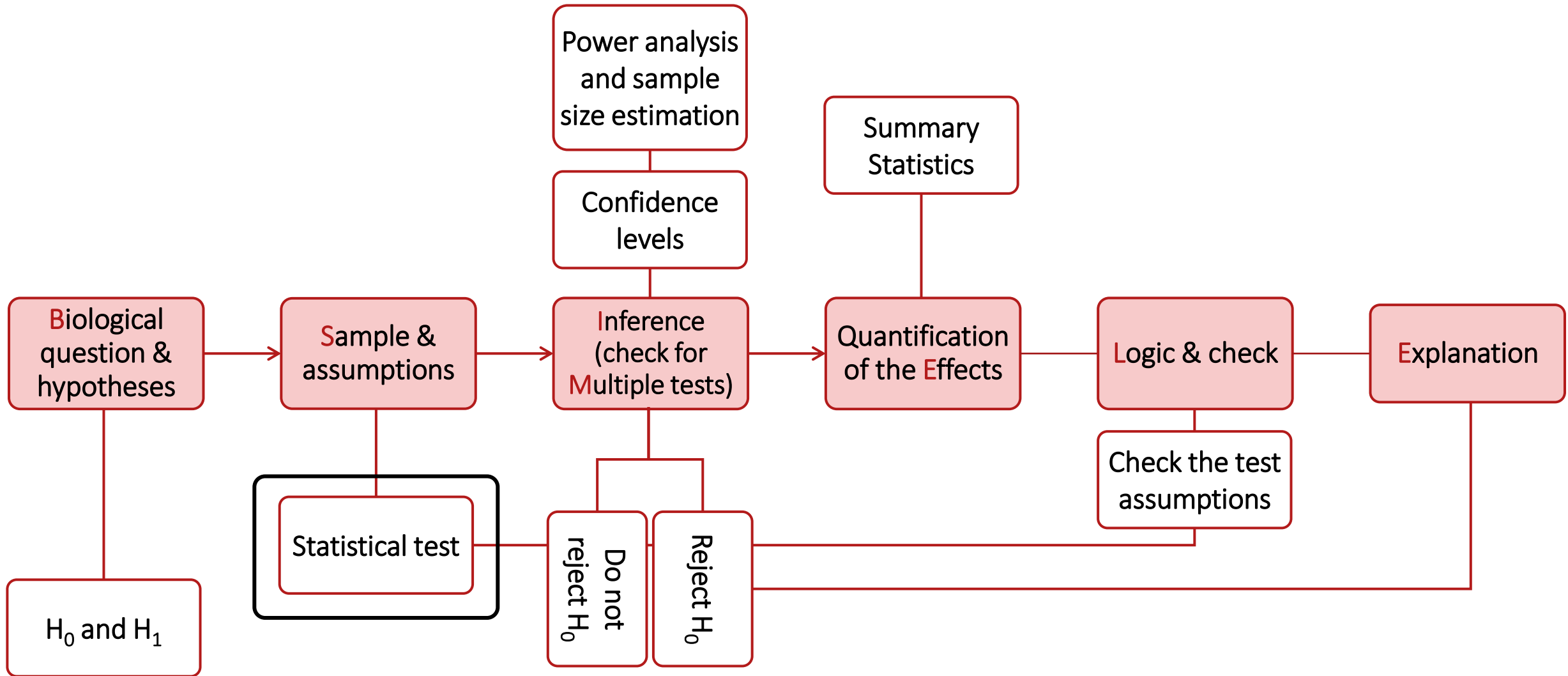
- **Homoscedasticity**: the variances of the subpopulations of Y are all equal and denoted by σ^2

Assumptions are on the subpopulations of Y for each value of X

A bit more complicated to test...



Hypothesis Testing Workflow



Pearson's Correlation Coefficient: the Math

- The bivariate normal distribution of X and Y has five parameters: μ_x , μ_y , σ_x , σ_y and ρ with **ρ the population correlation coefficient**
- For relationships following the assumptions of **linearity, independence, normality and homoscedasticity**, we can calculate the **Pearson's correlation coefficient** of bivariate populations and samples to measure the strength and direction of the **linear** relationship between X and Y

For the population, $\rho = \text{average} \left[\frac{X - \mu_x}{\sigma_x} \frac{Y - \mu_y}{\sigma_y} \right]$

→ Assess whether X and Y vary together...

Pearson's Correlation Coefficient: the Math

- The bivariate normal distribution of X and Y has five parameters: μ_x , μ_y , σ_x , σ_y and ρ with **ρ the population correlation coefficient**
- For relationships following the assumptions of **linearity, independence, normality and homoscedasticity**, we can calculate the **Pearson's correlation coefficient** of bivariate populations and samples to measure the strength and direction of the **linear** relationship between X and Y

For the sample,
$$r = \frac{1}{n-1} \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} \right] = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

→ Assess whether X and Y vary together...

Pearson's Correlation Coefficient: the Math

- Properties of the correlation coefficient

$$r \in [-1; 1]$$

$r \rightarrow 1$ for strong positive associations

$r \rightarrow -1$ for strong negative associations

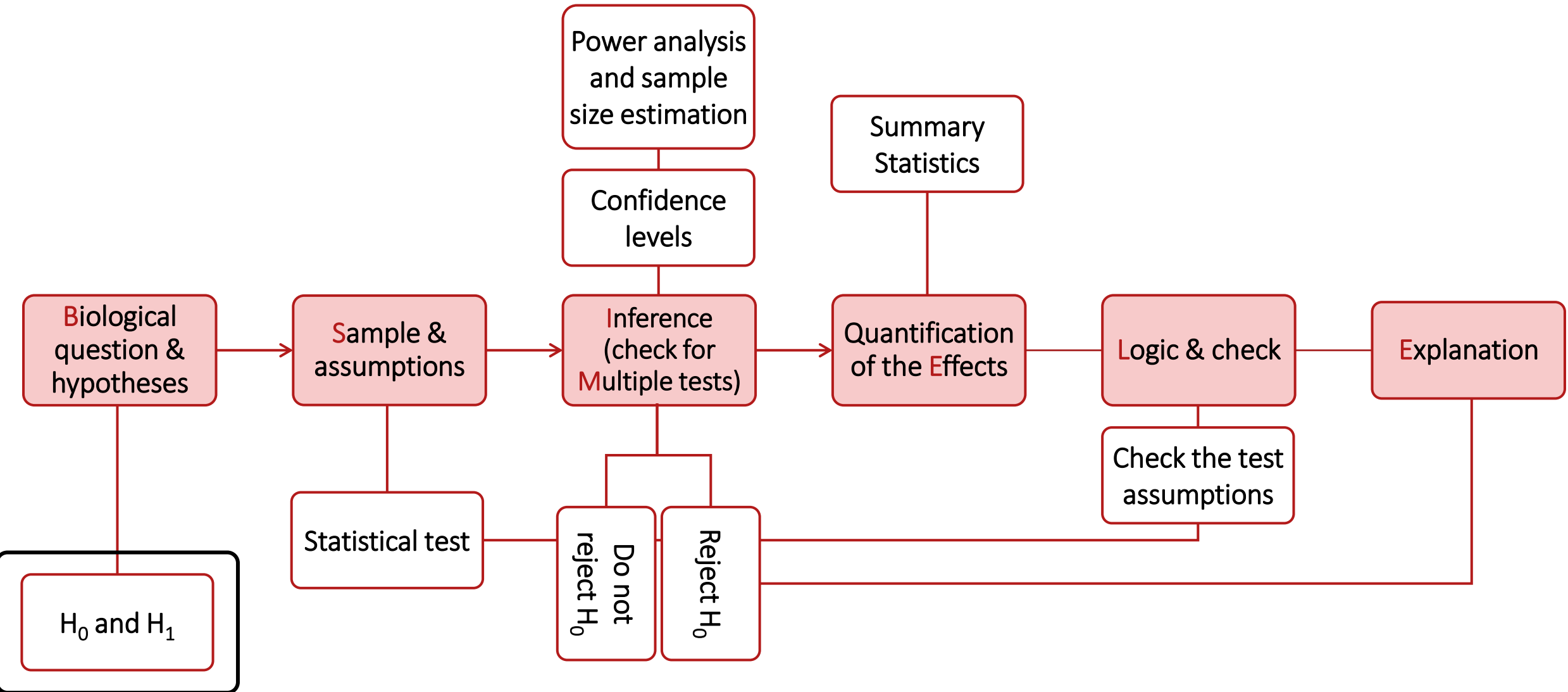
$r = 0$ in the absence of association

$r = 1$ for perfect associations

$$\text{For the sample, } r = \frac{1}{n-1} \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} \right] = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

→ Assess whether X and Y vary together...

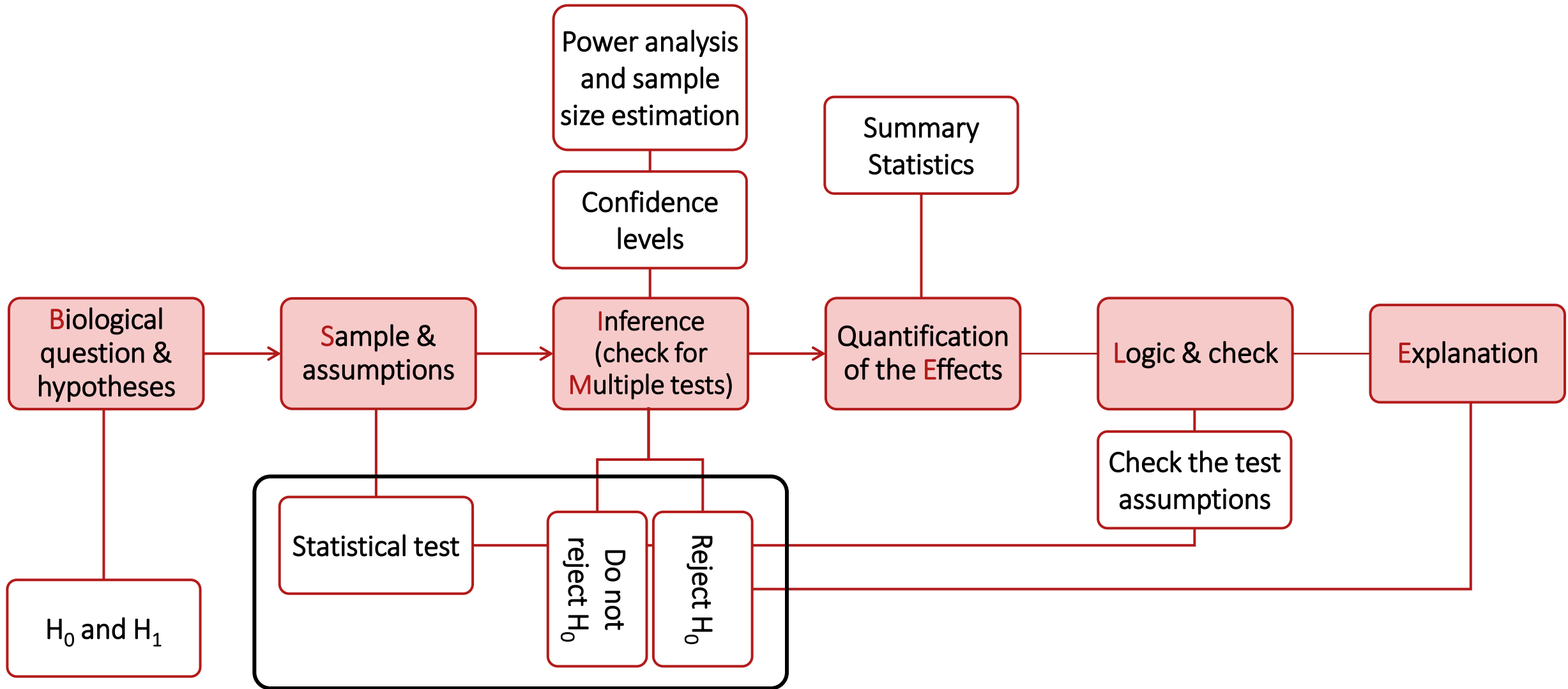
Hypothesis Testing Workflow



Working Hypotheses

- Null hypothesis (H_0): $\rho = 0$
 - There is no linear correlation (or association) between X and Y
 - X and Y are independent
- Alternative hypothesis (H_1): $\rho \neq 0$
 - There is a linear correlation (or association) between X and Y
 - X and Y are not independent

Hypothesis Testing Workflow



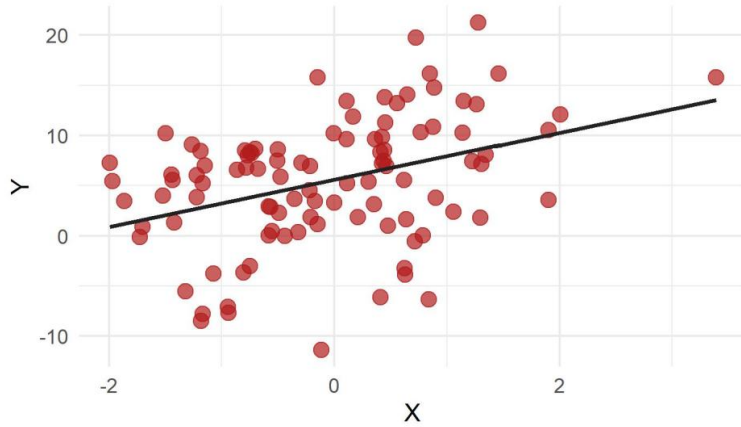
Pearson's Correlation Coefficient

- $r = \frac{1}{n-1} \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} \right] = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$, a descriptive parameter

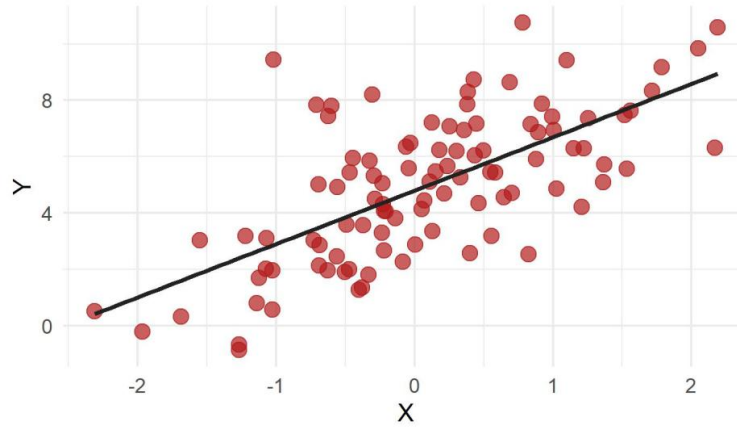
- $t = r \sqrt{\frac{n-2}{1-r^2}}$, the test statistics

With $n - 2$ degrees of freedom when exploring linear associations between 2 variables
To be compared to the known t -distribution (ttable.org), after having chosen the significance level $\alpha \rightarrow$ returns a p-value

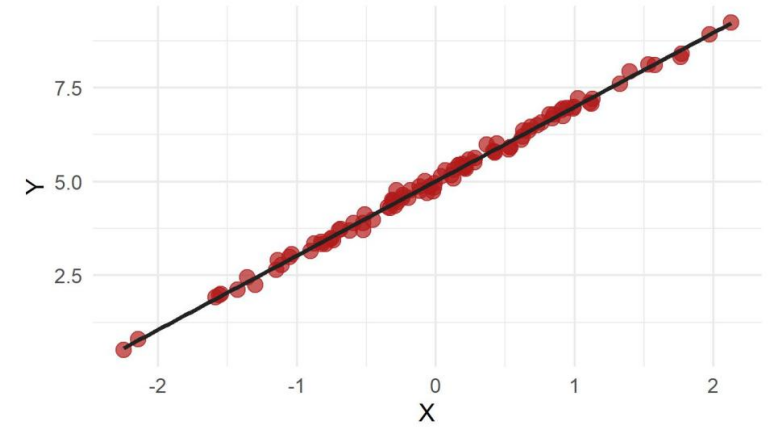
Pearson's Correlation Coefficient to Quantify Associations



Weak positive
correlation
 $r = 0.23$



So so positive
correlation
 $r = 0.67$



Strong positive
correlation
 $r = 0.99$

$$\text{With } r = \frac{1}{n-1} \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} \right] = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}},$$

the **sample correlation coefficient**, quantifying how X and Y vary together...

Testing & Quantifying Associations between Continuous Variables

Non-parametric equivalent

Which Statistical Test?

Context	Parametric Test	Non-Parametric Test
Comparison of 2 means	T-test	Wilcoxon rank test
Comparison of > 2 means	ANOVA	Kruskal-Wallis rank test
Correlation between 2 continuous variables	Pearson ($\pm$ linear regression)	Spearman rank test

Non-Parametric Regressions & Spearman's Correlation Coef.

- **Independence:** Y values are statistically independent (via ε_i)

$$y_i = a x_i + b + \varepsilon_i$$

- Works for non-linear monotonic correlations

- $r_s = 1 - \frac{6 \sum_{i=1}^n d^2}{n(n^2-1)}$ with d the rank difference

Testing & Quantifying Associations between Continuous Variables

Regression models basics with the linear regression

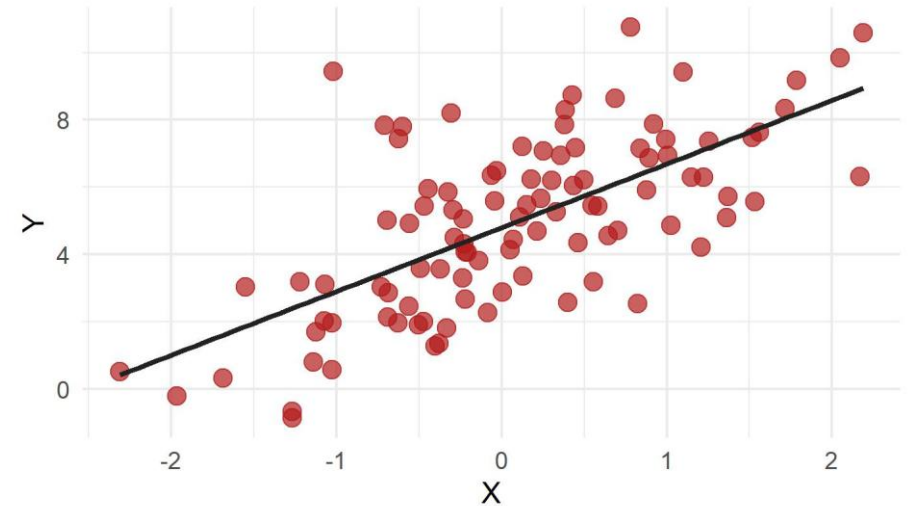
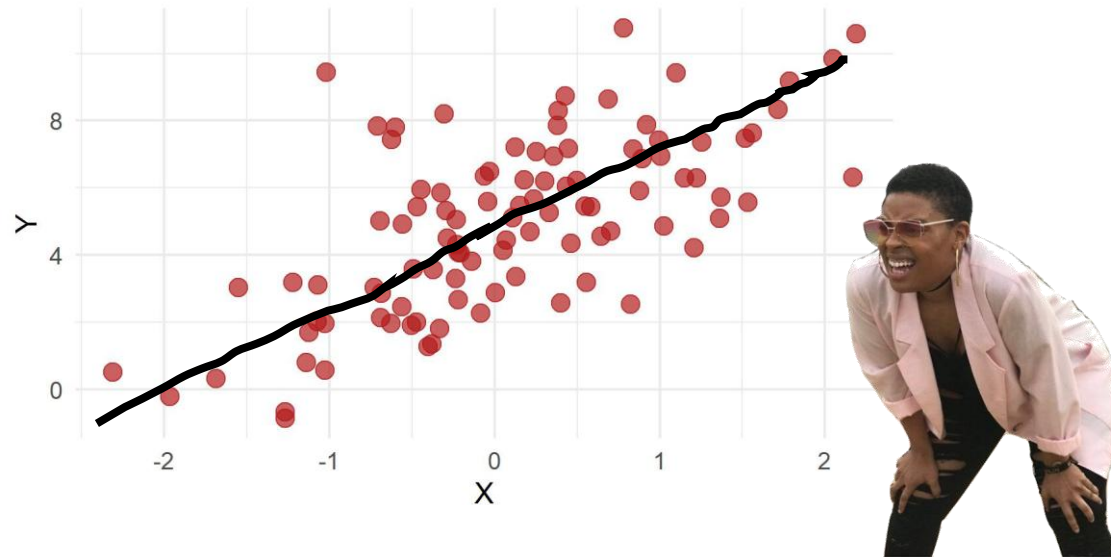
Linear Regressions

- Ok now we know there is a line...

What's the line though?

What are a and b in $Y = aX + b$?

With ordinary least square regressions



Linear Regressions

- Regression: **estimation** the relationships between variables

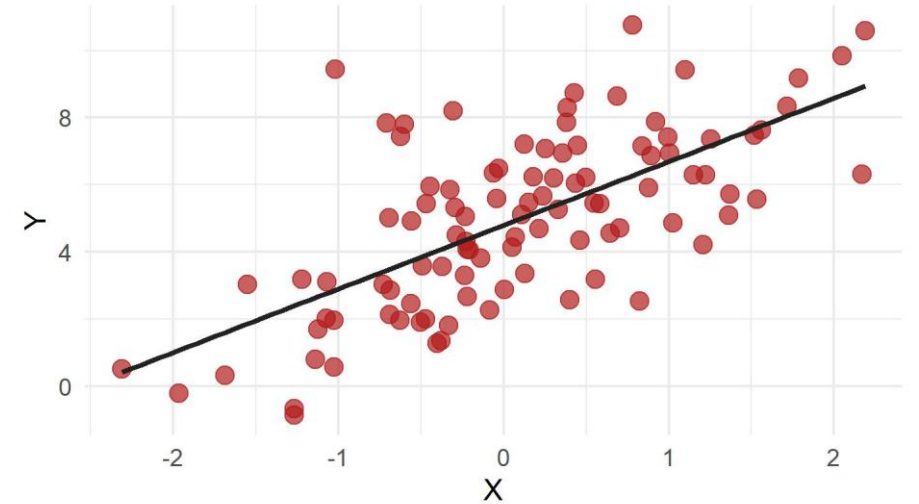
$$Y \sim X \text{ or } Y \sim X_1, X_2, \dots$$

→ An objective way to **predict** or estimate the value of one variable, given a value of another variable

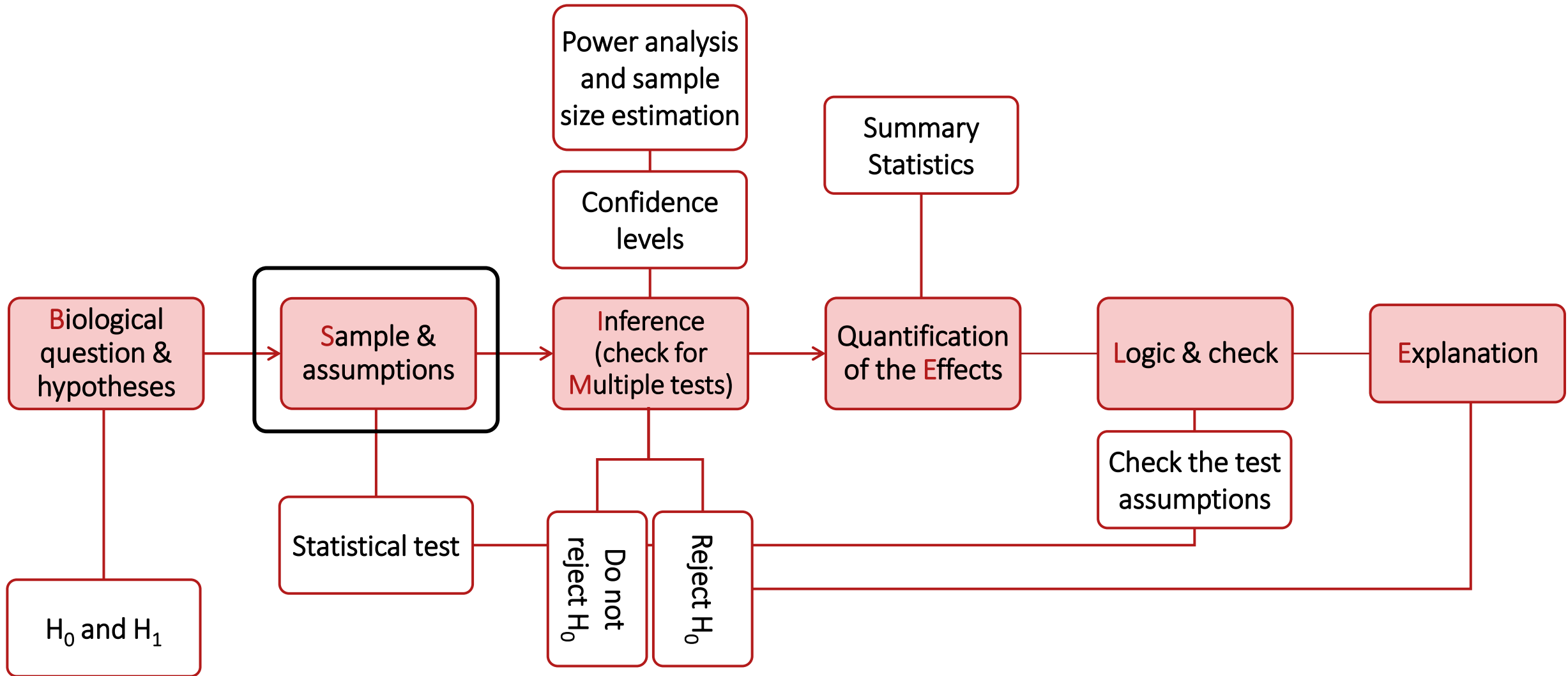
- **Linear** regression:

$$Y = aX + b \text{ and } y_i = \underbrace{ax_i + b}_{\text{Predicted value}} + \varepsilon_i$$

Individual variation,
or noise: deviance
from the predicted
value



Hypothesis Testing Workflow



Linear Regressions: Assumptions

- **Independence:** Y values are statistically independent (via ε_i)

$$y_i = a x_i + b + \varepsilon_i$$

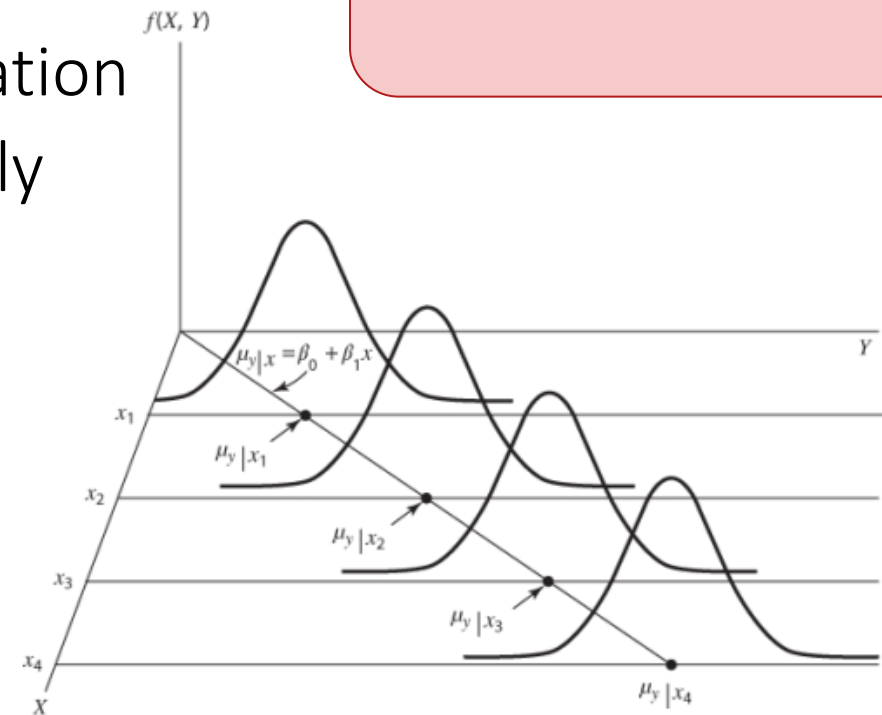
- **Linearity between X and Y :** the means of the subpopulations of Y all lie on the same straight line defined by

$$\mu_{y|x} = a x + b$$

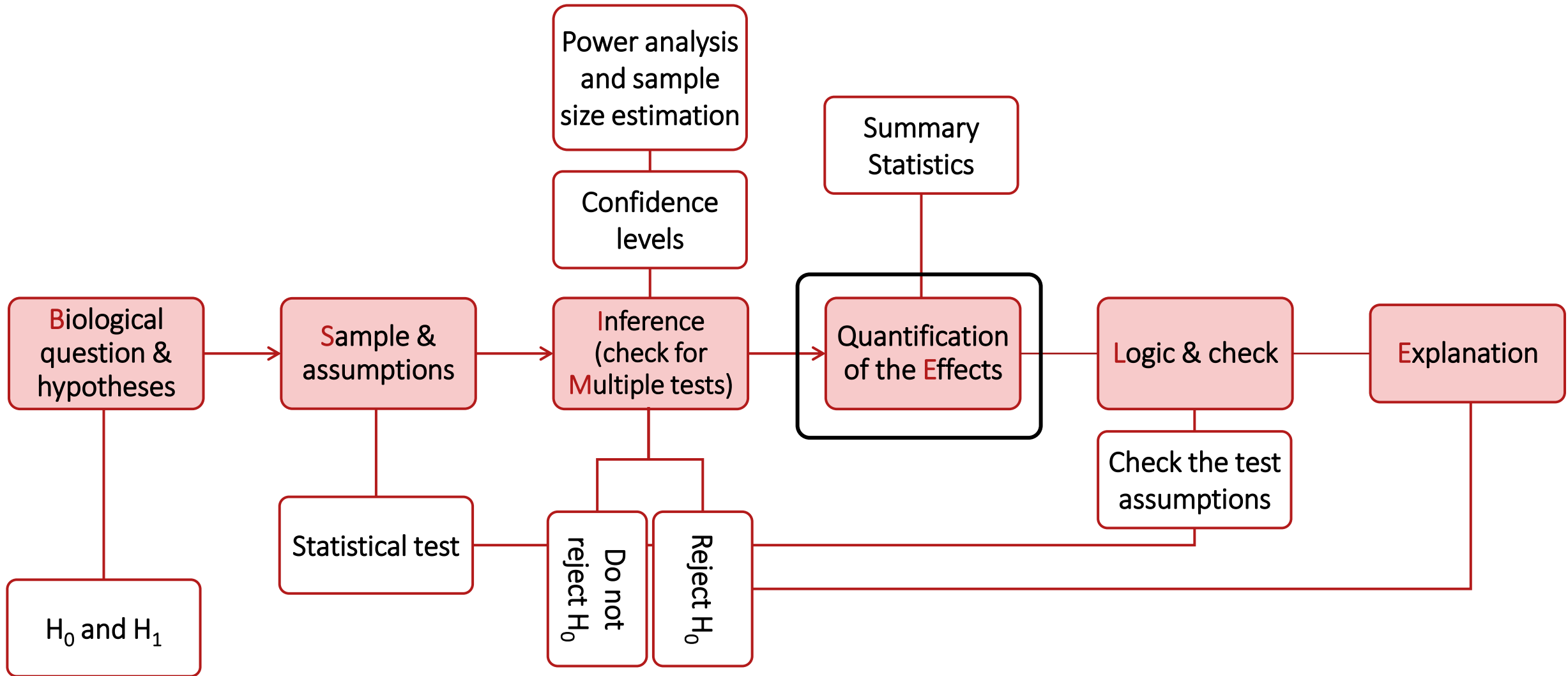
- **Normality:** for each value of X there is a subpopulation of Y values; these subpopulations must be normally distributed
- **Homoscedasticity:** the variances of the subpopulations of Y are all equal and denoted by σ^2

Assumptions are on the subpopulations of Y for each value of X

A bit more complicated to test...



Hypothesis Testing Workflow



Linear Regressions: the Math

- Regression: **estimation** the relationships between variables

$$Y \sim X \text{ or } Y \sim X_1, X_2, \dots$$

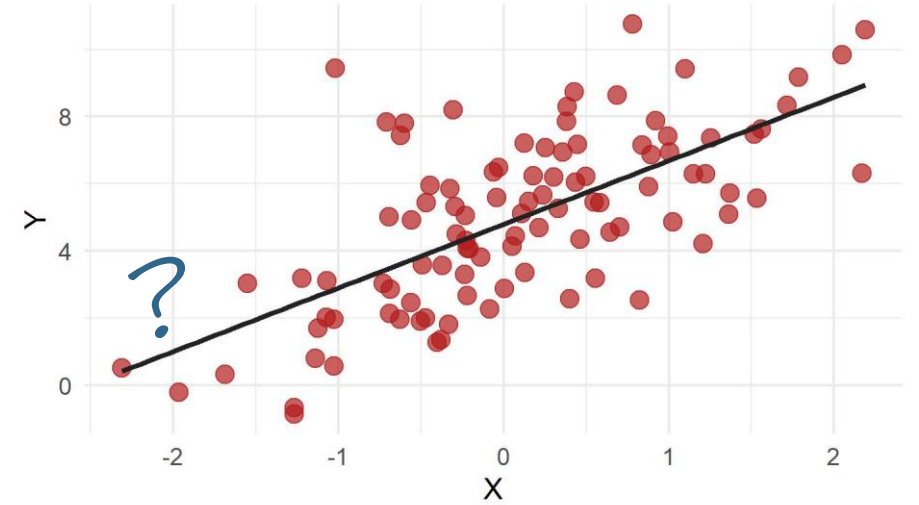
→ An objective way to **predict** or estimate the value of one variable, given a value of another variable

- Linear** regression:

$$Y = aX + b \text{ and } y_i = a x_i + b + \varepsilon_i$$

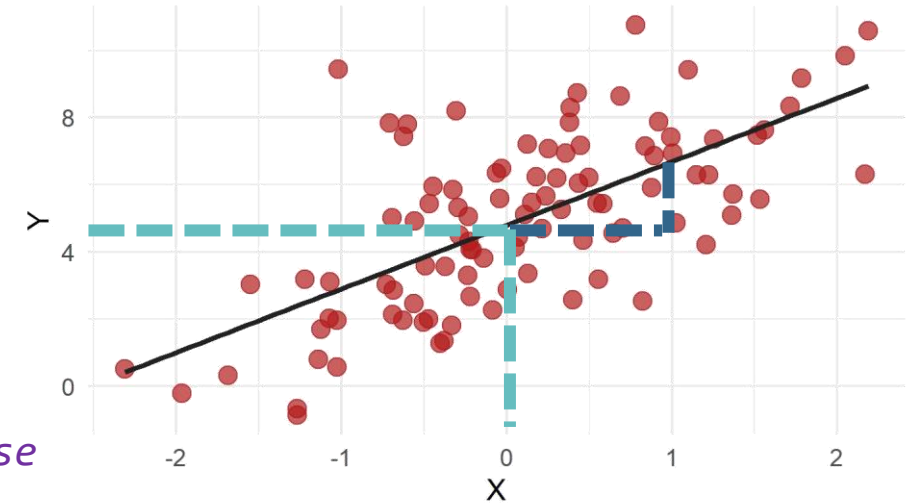
Predicted value ?

Individual variation,
or noise: deviance
from the predicted
value



Linear Regressions: the Math

```
# Define the parameters for the linear relationship
slope <- 2
intercept <- 5
n <- 100 # Number of data points
# Generate random values for x
x <- rnorm(n)
# Calculate corresponding y values using the linear relationship
y <- slope * x + intercept + rnorm(n, mean = 0, sd = 2) # Add noise
```



The effect size

The change in Y for a
one-unit change in X

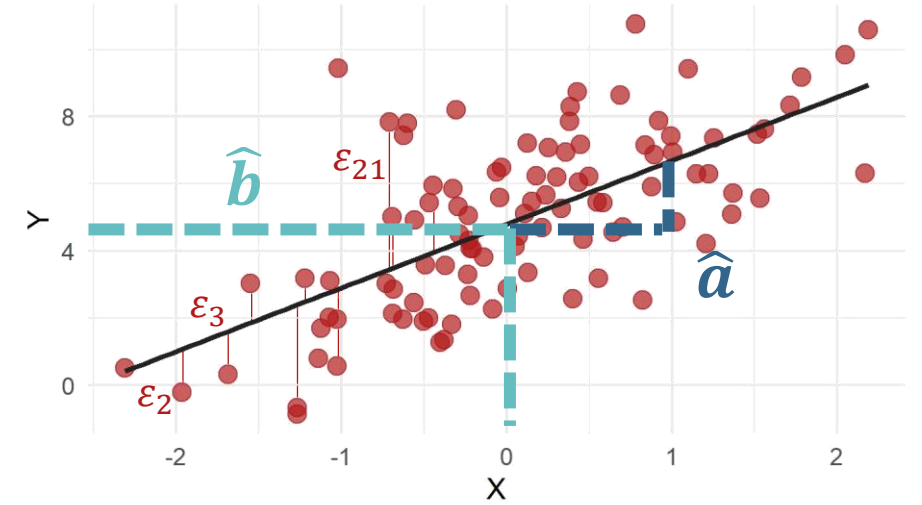
Slope

Intercept

$$Y = aX + b \text{ and } y_i = ax_i + b + \varepsilon_i$$

Linear Regressions: the Math

- Several methods for linear regressions (*i.e.*, to estimate a and b)
 - Ordinary least square (OLS)
 - Ridge regression
 - Lasso regression
 - ...



Minimize the sum of squared errors (*SSE*)

$$SSE = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Observed
(points)

Predicted
(line)

$$Y = aX + b \text{ and } y_i = ax_i + b + \epsilon_i$$

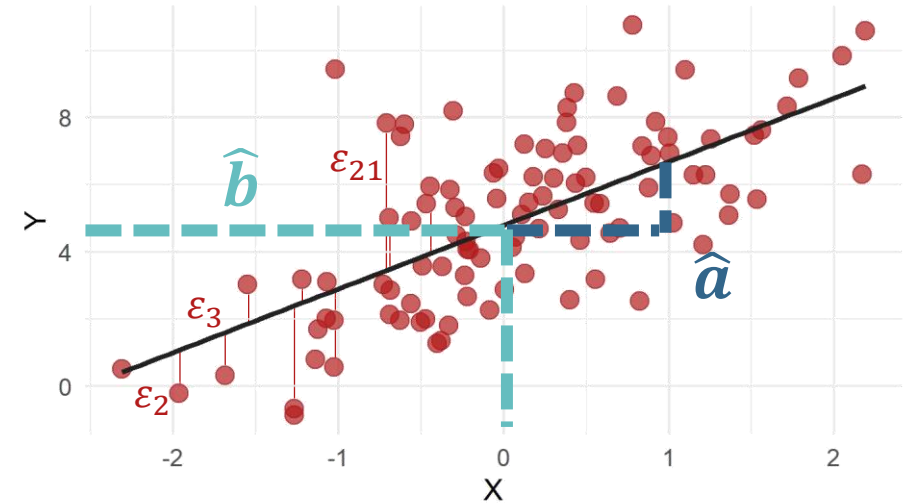
Linear Regressions: the Math

- We are looking for the values of a and b that minimize the $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ with $\hat{y}_i = \hat{a} x_i + \hat{b}$...

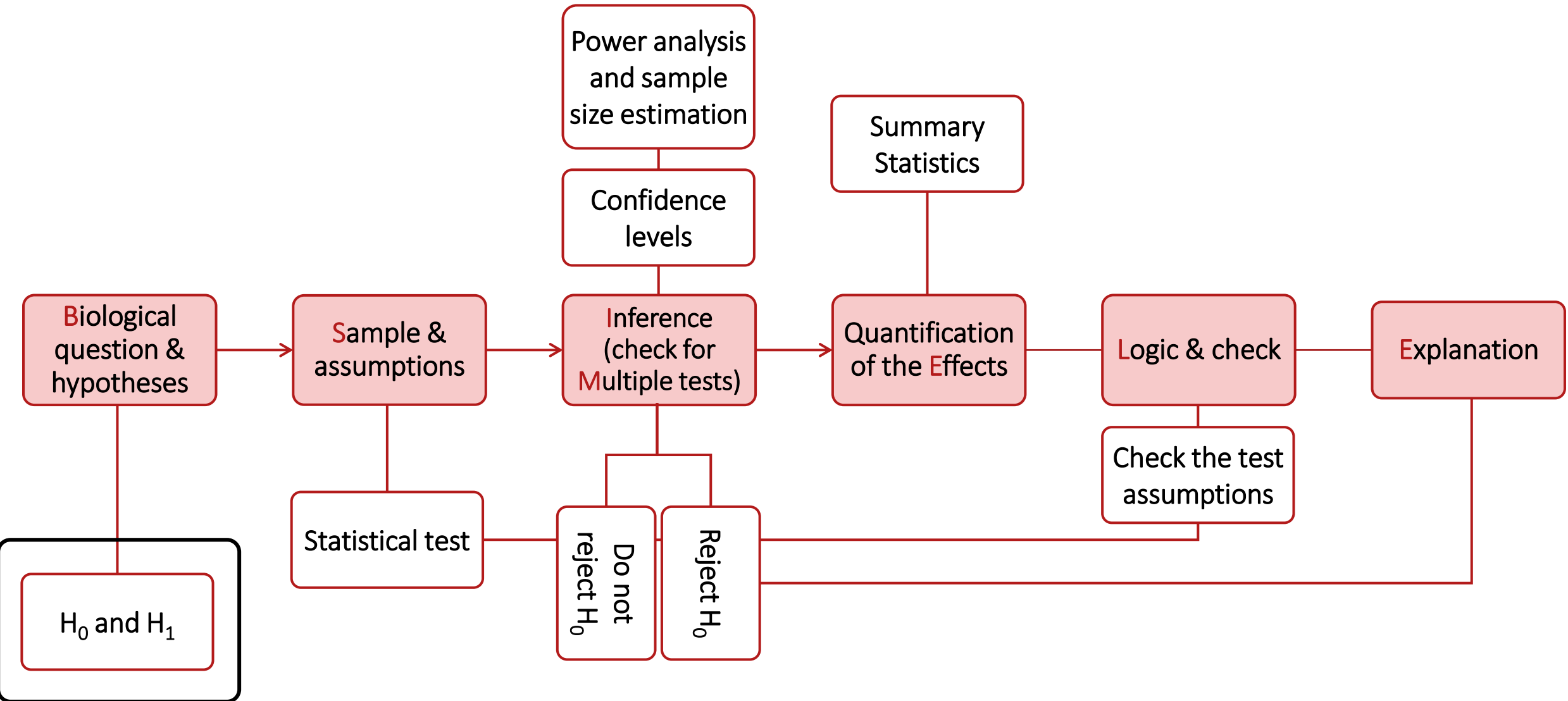
$$\hat{a} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{b} = \bar{y} - \hat{a}\bar{x}$$

The values for which the partial derivatives of the SSE with respect to a and b is equal to 0



Hypothesis Testing Workflow



Linear Regressions: Additional Hypothesis and Test

- Testing for the absence of association between 2 quantitative variables...
 - $H_0: \rho = 0$ and $H_1: \rho \neq 0$, with ρ the **true population correlation coef.**
 - $H_0: a = 0$ and $H_1: a \neq 0$, with a the **slope of the linear relationship** between the 2 variables

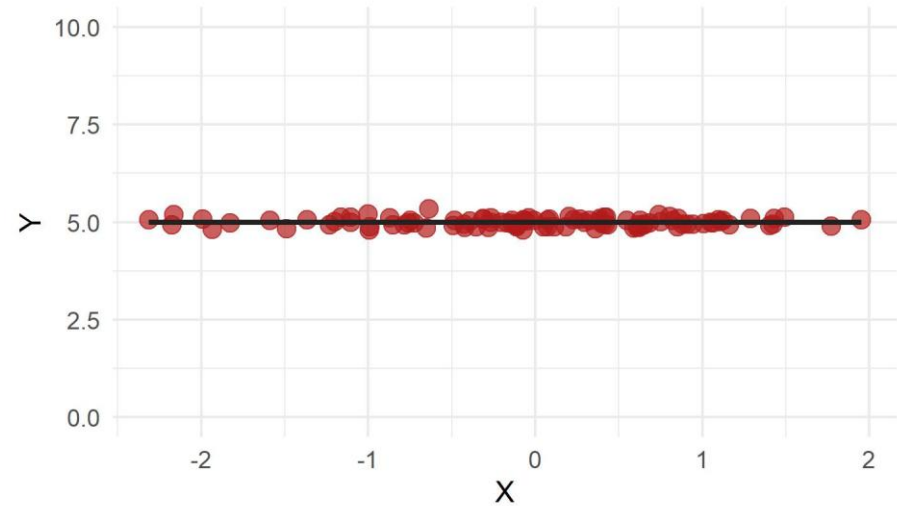
Alternative method useful for multivariate models (stay tuned...)

- Test on a ...
 - Point estimate for a ✓
 - Confidence interval...

- $\hat{a} = \hat{a} \pm t_{\alpha/2} \times s_{\hat{a}}$

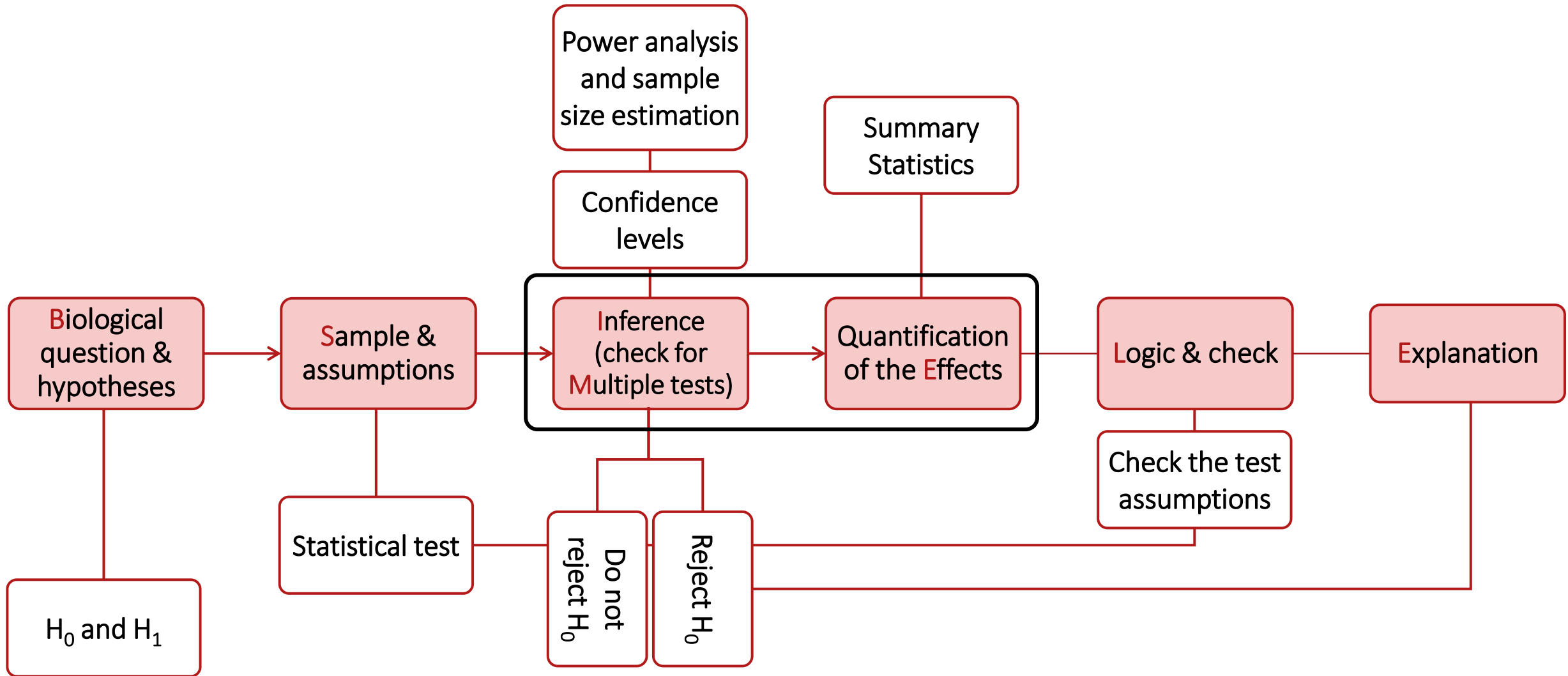
with $s_{\hat{a}} = \sqrt{\frac{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}{s_x \sqrt{n-1}}}$ ← SSE

and $s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$



A linear relationship with $a = 0$

Hypothesis Testing Workflow



Linear Regressions: Model Assessment

- Pearson's $r = \frac{1}{n-1} \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y} \right] = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}},$

the **sample correlation coefficient**, quantifying how X and Y vary together...

- OLS $r = \frac{s_x}{s_y} \hat{a}$

Or... $R^2 = \frac{SSY - SSE}{SSY}$ with $R^2 = r^2$ the **model coefficient of determination**

Assessing how much of the change in y is explained by the change in x (*i.e.*, how good the model $Y \sim X$ is good at predicting Y)

$SSY = \sum_{i=1}^n (y_i - \bar{y})^2$ the sum of squares (observations from the average)

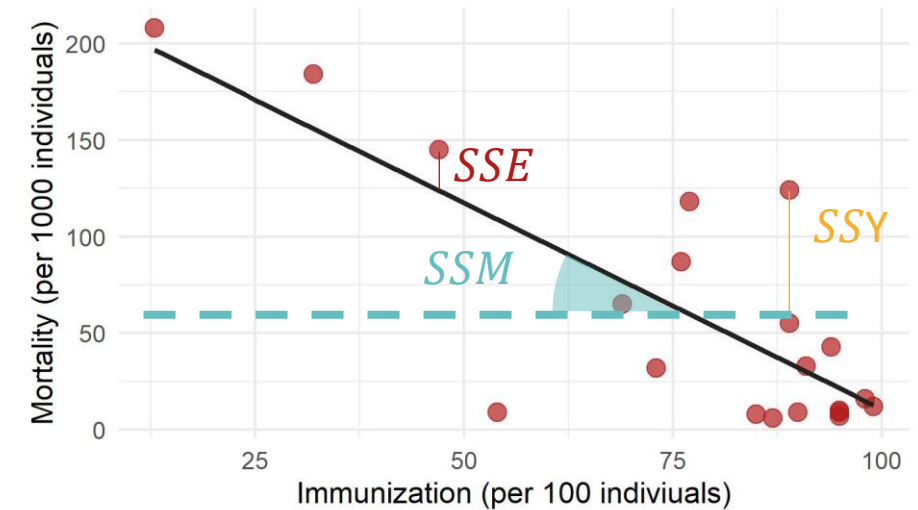
$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ the sum of squares of errors (obs. from the predictions)

Linear Regressions: Model Assessment

- $SSY = SSE + SSM$

Total variation is a sum of the distance of the model from the observed average, and individual observations from the model

≈ ANOVAs, $SS_{total} = SS_{between} + SS_{within}$



$SSY = \sum_{i=1}^n (y_i - \bar{y})^2$ the sum of squares (observations from the average)

$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ the sum of squares of errors (obs. from the predictions)

$SSM = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ the sum of squares of the model (pred. from the average)

Practice Time

Before Next Time (Friday March 13, in LH3)

- Practice: **Homework #07** due by **Thursday 5 PM**
- Use the Guiding Quiz #07 if needed
- Check you are ready for next class: **Knowledge Check #07** due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours (in-person or zoom)
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Madison)
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Daniel, Harsh)
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - in S1-122, unless indicated otherwise on Canvas. Zoom link on syllabus.